Overview

The firmware controls a system consisting of multiple DC motors and sensors. It interprets and executes serial commands to adjust motor positions, motor speeds, sensor sensitivities, and to select or execute predefined motor position arrays. Additionally, it incorporates an analog control mechanism using inputs from two variable resistors for interactive array control.

Motor Control and Arrays

- Each motor (Motor 1 to Motor 6) can be individually controlled to move to a specific position or at a certain speed.
- Predefined arrays (A to G) represent specific configurations of all motor positions. For example, selecting Array A will position Motor 1 at 700, Motor 6 at 700, and so on, while other motors (2, 3, 4) will be at zero.
- The user can excute an array which actively moves motors to the set positions.
- The motor speed command adjusts how quickly motors move to their target positions.

Sensor Sensitivity

- Each sensor's sensitivity can be adjusted via serial commands, allowing customization of the system's responsiveness to external inputs.

Demo Mode

- In Demo Mode, the system automatically cycles through each array (A to F), with a one-second pause between each.
- Between each array, Array G is executed, ensuring a neutral starting position for each new array.

Safety and Feedback (Serial Monitor)

- The system includes safety checks and error handling to prevent malfunction or damage.
- Feedback is provided after executing commands or changing settings, ensuring the user is informed about the system's status.

Usage Scenario

- A user can input a command, like **SET MOTOR 1 POSITION 500**, to move Motor 1 to position 500.
- By sending **START DEMO**, the system showcases its capabilities by cycling through each array.
- Adjusting the attached variable resistors can change the active array or cycle through arrays without needing additional commands.

This system, with its combination of precise control, adjustable settings, and interactive elements, offers a versatile platform for various applications, while the variable resistors provide a unique and user-friendly way to interact with the system.

Variable Resistors for Control:

1. **General Functionality:**
    - Two analog inputs, variable resistors, are used for interactive control of the system.
2. **Resistor 1 (Array Activation/Deactivation Control):**
    - Users can set an activation value for Resistor 1 through a specific command.
    - When the value read from Resistor 1 equals or exceeds this activation value, the currently selected array is activated (motors move to their respective positions as defined in the array).
    - Once activated, the array remains active until the value read from Resistor 1 drops below the activation value minus 100 (hysteresis). This deactivation prevents rapid toggling.
3. **Resistor 2 (Array Cycling Control):**
    - An activation value can also be set for Resistor 2.
    - When the value from Resistor 2 meets or exceeds this threshold, the system cycles to the next array.
    - After cycling, the array will not change again until Resistor 2's value falls below the activation threshold minus 100 and then exceeds the activation threshold again. This mechanism prevents continuous cycling with sustained high values.
4. **Application and Interaction:**
    - These controls allow for an intuitive and dynamic way to manipulate the system without using serial commands, adding a layer of physical interaction.

Motor Control: Arrays Table

Array	Motor 6 Pos	Motor 6 Speed	Motor 5 Pos	Motor 5 Speed	Motor 1 Pos	Motor 1 Speed	Motor 2 Pos	Motor 2 Speed	Motor 3 Pos	Motor 3 Speed	Motor 4 Pos	Motor 4 Speed
A	700	255	500	255	700	255	0	255	0	255	0	255
B	700	255	1000	255	700	255	700	255	0	255	0	255
C	900	255	900	255	900	255	900	255	900	255	900	255
D	0	255	500	255	900	255	900	255	900	255	900	255
E	800	255	500	255	500	255	0	255	0	255	0	255
F	700	255	700	255	850	255	850	255	850	255	850	255
G	0	255	0	255	0	255	0	255	0	255	0	255

*255 is the default motor speed, the user can change the default motor speed as well

Command Structure Table

Command Type	Command Format	Description	Example
Set Individual Motor Position	`SET MOTOR <Motor Number> POSITION <Value>`	Sets the position of a specified motor.	`SET MOTOR 1 POSITION 700`
Set Motor Speed	`SET MOTOR <Motor Number> SPEED <Value>`	Adjusts the speed of a specified motor. Speed range: 100-255.	`SET MOTOR 1 SPEED 150`
Set Sensor Sensitivity	`SET SENSOR <Sensor Number> SENSITIVITY <Value>`	Sets the sensitivity for a specified sensor. Sensitivity range: 1-10.	`SET SENSOR 2 SENSITIVITY 5`
Execute Predefined Array	`<Array Letter>`	Executes the movements to achieve the motor positions defined in the chosen array.	`A`
Query Motor Speed	`QUERY MOTOR <Motor Number> SPEED`	Retrieves the current speed setting of a specified motor.	`QUERY MOTOR 1 SPEED`

Query Sensor Sensitivity	**QUERY SENSOR <Sensor Number> SENSITIVITY**	Retrieves the current sensitivity setting of a specified sensor.	**QUERY SENSOR 2 SENSITIVITY**
Save Current Configuration	**SAVE CONFIG**	Saves the current configuration of motor positions, speeds, and sensor sensitivities.	**SAVE CONFIG**
Load Saved Configuration	**LOAD CONFIG**	Loads the most recently saved configuration of settings.	**LOAD CONFIG**
Start Demo Mode	**START DEMO**	Initiates a demonstration mode that cycles through Arrays A-F with Array G between each.	**START DEMO**
Stop Demo Mode	**STOP DEMO**	Stops the demonstration mode.	**STOP DEMO**
Set Resistor Activation Value	**SET R<Resistor Number>, <Value>**	Sets the activation threshold for a given resistor.	**SET R1,300**

System Overview

Hardware Components:

1. **Motors (6 in total):** Each motor controls a specific part of the system, capable of moving to positions between 0 and 1000.
2. **Force Sensors (6 in total):** Each sensor is linked to a corresponding motor. The sensor detects force and controls the motor's movement.
3. **Variable Resistors (2 in total):** Used for interactive control of the system as control signals.
4. **Encoders (6 in total)**: Each motor is equipped with an encoder to provide precise feedback on its position and movement.

Functionalities:

1. **Motor Position Control:**
    - Each motor's position can be individually set via commands.
    - Predefined arrays (A to G) represent specific motor position configurations for different operational modes.
2. **Motor Speed Control:**
    - The speed of each motor's movement can be adjusted (range 100-255).
3. **Force Sensor Interaction:**
    - Each motor is paired with a force sensor.
    - When a force sensor senses force at or above its set sensitivity level, the corresponding motor stops, regardless of whether it has reached its targeted position in the array.
4. **Variable Resistor Control:**
    - Resistor 1 activates/deactivates the current array based on a set threshold.
    - Resistor 2 cycles through arrays once its value crosses a predefined activation threshold.
5. **Demo Mode:**
    - Cycles through Arrays A-F sequentially with a 1-second delay, reverting to Array G between each array.
6. **Command Structure:**
    - Includes commands for setting motor positions, Zero Position, motor speeds, sensor sensitivities, array selection, execution, and system queries.

7. **Encoder Feedback:**
    - Encoders provide real-time feedback on the position and movement of each motor, enhancing precision and control.